THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 07

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(d)** $(-5, 7)$	reflection in the y-axis changes the sign of the x-coordinate
2	**(a)** $4n - 1$	common difference 4, so $a_n = 4n - 1$
3	**(c)** terminating	denominator 25 is of the form $2^a 5^b$
4	**(d)** $(7x + 2y)^2$	identify $a^2 + 2ab + b^2 = (a + b)^2$
5	**(b)** 5 cm	$d = \sqrt{13^2 - 12^2} = 5$
6	**(a)** 44 cm	$\frac{180}{360} \times 2\pi r = 44$
7	**(a)** Jñān-Chaupaḍ	as given in the chapter
8	**(b)** 1275	$\frac{50 \cdot 51}{2}$
9	**(c)** either true or false	definition
10	**(a)** both true	adding 5 can be undone by subtracting 5
11	**(d)** 3 litres	12×10% = 1.2 litres of solute; total volume = 15 litres
12	**(c)** 81	$\frac{12 \times 85 + 33}{13} = 81$
13	**(b)** 2	carries by column: [1, 1, 0]
14	**(b)** 11	values 2, 3, …, min(m, n)
15	**(b)** Both A and R are true but R is not the correct explanation of A	both true, but concurrency does not explain the 2 : 1 ratio — the Midpoint Theorem does
16	**(d)** 24 cm	each side of the medial triangle is half of a side
17	**(b)** 1	k/3 = 2/6 gives k = 1 and then 5/12 ≠ 1/3
18	**(c)** 616 cm^2	$4\pi r^2 = 616$
19	**(d)** 248 cm^2	$2(lw + wh + lh) = 2(60 + 24 + 40) = 248$
20	**(d)** A is false but R is true	SA becomes four times; A is false

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$O(0,0),\ A(7,0),\ B(7,4),\ C(0,4)$	2
22	The square with side 10 units; area 100 sq. units (e.g. $9 \times 11 = 99 < 100$).	1+1
23	7 months (Jan, Mar, May, Jul, Aug, Oct, Dec) out of 12: $\frac{7}{12}$.	1+1
24	₹240, ₹280, ₹320 — an AP with $d = 40$ (1). $t_n = 200 + 40n$; $t_{10} =$ ₹600 (1).	1+1
25	Converse: If n has exactly 3 factors, then n is the square of a prime (1). It is true: three factors means $1, p, p^2$ for a prime p (1). (No counterexample exists.)	1+1
26	The total of all the components (1), so the chart helps compare totals while keeping track of the parts (1).	1+1
27	The sum is 370° (1), but the angle sum of a quadrilateral is 360°, so it cannot (1).	1+1
28	Substitute $x = 1$, $y = k$: $3 + 2k = 7$ (1). So $k = 2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) 2, 3, 5, 7 ⇒ $\frac{4}{8} = \frac{1}{2}$ (ii) 6, 7, 8 ⇒ $\frac{3}{8}$.	1+1+1
30	Examine the multiples $m, 2m, 3m, \ldots$ one by one (1). Stop at the first multiple that is divisible by n (1). Report that multiple as the lcm (1).	1+1+1
31	$lw \cdot wh \cdot lh = (lwh)^2 = 12 \times 20 \times 15 = 3600$ (2). So $V = lwh = 60$ cm^3 (1).	2+1
32	$b(x) = 600 - 15x$; zero when $x = 40$ days. Negative slope ⇒ linear decay.	2+1
33	$18,\ 13.5,\ 10.125$ ft (2). It is a GP with $r = \frac{3}{4}$ (1).	2+1
34	$x(x - 4) = 96 \Rightarrow x^2 - 4x - 96 = 0 \Rightarrow (x - 12)(x + 8) = 0$. $x = 12$ m (reject −8); breadth 8 m.	1+1+1

35	(i) $212 = \frac{9}{5}C + 32$, so $\frac{9}{5}C = 180$ and $C = 100$ °C (1). (ii) Put $F = C$: $C = \frac{9}{5}C + 32$, so $-\frac{4}{5}C = 32$ (1). $C = -40$; the readings are equal at -40 (1).	1+1+1
	OR: Multiply by 3: $3y = 2x - 12$, so $2x - 3y - 12 = 0$ (1). At $(6, 0)$: $12 - 0 - 12 = 0$ (1). At $(3, -2)$: $6 + 6 - 12 = 0$ (1).	1+1+1
36	Converse: If a triangle has two equal angles, then it is isosceles (1). It is true (1). Draw the altitude from the third vertex; the two right triangles are congruent by AAS, so the sides opposite the equal angles are equal (1).	1+1+1
	OR: Converse: If n is not a multiple of 3, then n and $n + 3$ have no common factor (1). Their common factors divide the difference 3, so only 1 or 3 can be common (1). If n is not a multiple of 3, 3 does not divide n, so the converse is true (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Option 1: $\frac{80+60+70}{3} = 70$ (1); Option 2: $\frac{80+120+350}{8} = \frac{550}{8} = 68.75$ (1½). (ii) Option 2 gives more weight to the exam; for a student scoring higher in the exam it is better (1½). (iii) Weighted mean (1).	2½+1½+1
	OR: (i) 2015 total 1000: 60%, 10%, 20%, 10% (1½). 2025 total 2000: 40%, 30%, 20%, 10% (2). (ii) Solar's share rose from 10% to 30% and in MW from 100 to 600, a sixfold rise, the fastest growth (1½).	3½+1½
38	(i)(a) $225 + 400 = 625 = 25^2$ ✓ right-angled (1½). (b) $121 + 3600 = 3721$; $62^2 = 3844$ × not right-angled (1½). (ii) Build a right triangle with legs a, b; its hypotenuse is $\sqrt{a^2 + b^2} = c$; both triangles have the same three sides, so they are congruent by SSS and the given one has a right angle (2).	3+2
	OR: (i) If a quadrilateral is a rectangle, then its diagonals are equal (1). (ii) If a quadrilateral has equal diagonals, then it is a rectangle (1). (iii) An isosceles trapezium with parallel sides 6 cm and 10 cm and legs 5 cm has equal diagonals but no right angle (1½). (iv) A parallelogram: if a parallelogram has equal diagonals, it is a rectangle (1½).	1+1+1½+1½
39	Let $P(x, 0)$: $(x + 2)^2 + 25 = (x - 2)^2 + 9 \Rightarrow 8x = -16 \Rightarrow x = -2$. $P(-2, 0)$; distance $= \sqrt{0 + 25} = 5$.	3+2
	OR: $(k - 3)^2 + 36 = 100 \Rightarrow (k - 3)^2 = 64 \Rightarrow k = 11$ or $k = -5$. Points $(11, 2)$ and $(-5, 2)$.	3+2
40	(i) Primes 2,3,5,7 ⇒ $\frac{4}{8} = \frac{1}{2}$ (1). (ii) Composites 4,6,8 ⇒ $\frac{3}{8}$ (1). (iii) Not fair, $\frac{1}{2} \neq \frac{3}{8}$ (1). (iv) 1 (neither prime nor composite), probability $\frac{1}{8}$ (2).	1+1+1+2
	OR: (i) $\frac{13}{52} = \frac{1}{4}$ (1). (ii) $\frac{39}{52} = \frac{3}{4} = 1 - \frac{1}{4}$ (2). (iii) $\frac{26}{52} = \frac{1}{2}$ (1). (iv) 8 cards ⇒ $\frac{8}{52} = \frac{2}{13}$ (1).	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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